SOFTWARE REQUIREMENTS SPECIFICATION

for

MIRRORWALK

A Native Android Application for Spatially Anchored Ghost-Race Running and Route Tracking

Version 1.0

Prepared by:
1. Ana Hasan (1024160103)
2. Agrim Gupta (1024160117)
3. Janvi Verma (1024160108)
4. Yashkaran Singh (1024160125)

Group: 9, BE Third Year, CSE

Submitted to: Anushka Ma'am

Computer Science and Engineering Department
Thapar Institute of Engineering and Technology, Patiala

October 16, 2026

Contents

1 Introduction **5**
- 1.1 Purpose . . . 5
- 1.2 Document Conventions . . . 5
- 1.3 Intended Audience and Reading Suggestions . . . 5
- 1.4 Project Scope . . . 6
- 1.5 Definitions, Acronyms and Abbreviations . . . 6
- 1.6 Entire Workflow of MirrorWalk . . . 7
- 1.7 References . . . 7

2 Overall Description **8**
- 2.1 Product Perspective . . . 8
- 2.2 Product Functions . . . 8
- 2.3 Data Model . . . 9
- 2.4 User Classes and Characteristics . . . 9
- 2.5 Operating Environment . . . 10
- 2.6 Design and Implementation Constraints . . . 11
- 2.7 Assumptions and Dependencies . . . 11
- 2.8 User Activities . . . 11
  - 2.8.1 Runner Activities . . . 11
  - 2.8.2 Ghost Runner Activities . . . 11
  - 2.8.3 Social Activities (Planned) . . . 13

3 External Interface Requirements **14**
- 3.1 User Interfaces . . . 14
- 3.2 Hardware Interfaces . . . 14
- 3.3 Software Interfaces . . . 14
- 3.4 Communications Interfaces . . . 15

4 System Features **16**
- 4.1 Run/Walk Session Control . . . 16
- 4.2 Location Permission . . . 16
- 4.3 GPS Tracking and Filtering . . . 16
- 4.4 Live Statistics . . . 17
- 4.5 Route Display . . . 17
- 4.6 Saving Runs . . . 17
- 4.7 Run History . . . 17
- 4.8 Ghost Mode and AR Ghost Runner . . . 17

4.9 Live Comparison Overlay 18
4.10 Social Features 18
4.11 Fitness Analysis 18

5 Other Nonfunctional Requirements **19**
5.1 Performance Requirements 19
5.2 Safety Requirements 19
5.3 Security and Privacy Requirements 19
5.4 Software Quality Attributes 19
5.5 Business Rules 20

6 Other Requirements **21**
6.1 Requirements Status Summary 21
6.2 Maintenance and Future Changes 21

References **21**

1 Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for **MirrorWalk**, a native Android application for runners and walkers. It describes what the system must do, the limits within which it must work, and the quality it must achieve. It covers release 1.0, which is the current prototype together with the features that are planned for later releases.

MirrorWalk combines three ideas in one app: (1) accurate live GPS-based run tracking, (2) an Instagram-style social layer built around running, and (3) an augmented-reality (AR) *ghost runner* that lets a user race against the recording of a previous run, shown through the phone camera. Most fitness apps only show numbers and most social apps do not track runs, so MirrorWalk joins both worlds and adds a visual, motivating way to compete against past performance.

1.2 Document Conventions

Every requirement in this document is marked with a status and a priority:

- **[Implemented]** – already working in the current prototype.
- **[Planned]** – designed, but not built yet.
- **High / Medium / Low** – priority of the feature for the success of the product.

Functional requirements have identifiers such as `FR-3.2`, and non-functional requirements have identifiers such as `NFR-2`. The word "shall" means a mandatory requirement.

1.3 Intended Audience and Reading Suggestions

This SRS is written for developers, project guides and evaluators, testers, and future maintainers of MirrorWalk.

- **Evaluators and guides** should read Chapters 1, 2 and the summary table in Chapter 6.
- **Developers** should read Chapters 2, 3 and 4.
- **Testers** should read Chapter 4 (functional requirements) and Chapter 5 (quality requirements).

1.4 Project Scope

MirrorWalk is a mobile app that lets a user start, pause, resume and stop a run or walk session, records the route using GPS, calculates live distance, pace and time, saves completed runs on the device, and allows the user to race against a "ghost" of a previous run in AR.

In scope for release 1.0 (prototype):

- Run/walk session control and live statistics.
- GPS tracking with filtering of bad points.
- Live and final route display on a map.
- Local saving, viewing and deleting of runs.
- Ghost mode with an AR ghost runner (using hardcoded sample data, structured so it can be replaced by a saved run later).

Planned for later releases:

- User sign up and login, user profiles, followers and following counts.
- Social feed with posts of runs and optional run photos.
- Fitness analysis (trends of pace, distance and progress).
- Using any real saved run (own or a friend's) as a ghost.

1.5 Definitions, Acronyms and Abbreviations

Term	**Meaning**
AR	Augmented Reality: virtual objects shown on top of the real world through the camera.
ARCore	Google's framework for AR on Android phones.
Ghost runner	A virtual runner that replays a recorded run so the user can race against it.
GPS	Global Positioning System, used to find the user's location.
Fused Location Provider	Google Play Services API that combines GPS, Wi-Fi and mobile network data to give the device location.
Pace	Time taken to cover one kilometre (minutes per km).
Plane detection	ARCore's ability to find flat surfaces such as the ground in the camera view.
Room	Android library used to store data in a local database.
MapLibre	Open-source map rendering library for Android.

Term	Meaning
OpenFreeMap	Free source of map tiles that needs no API key.
ViewModel	Android component that holds the state of a screen.
SRS	Software Requirements Specification.

1.6 Entire Workflow of MirrorWalk

Figure 1.1 gives an overview of the project. It shows how the main entities depend on each other: the Runner starts a Run Session, location services track it, and it becomes a Run Record that is stored in the Room database and can be replayed as an AR Ghost Runner that the Runner races against.

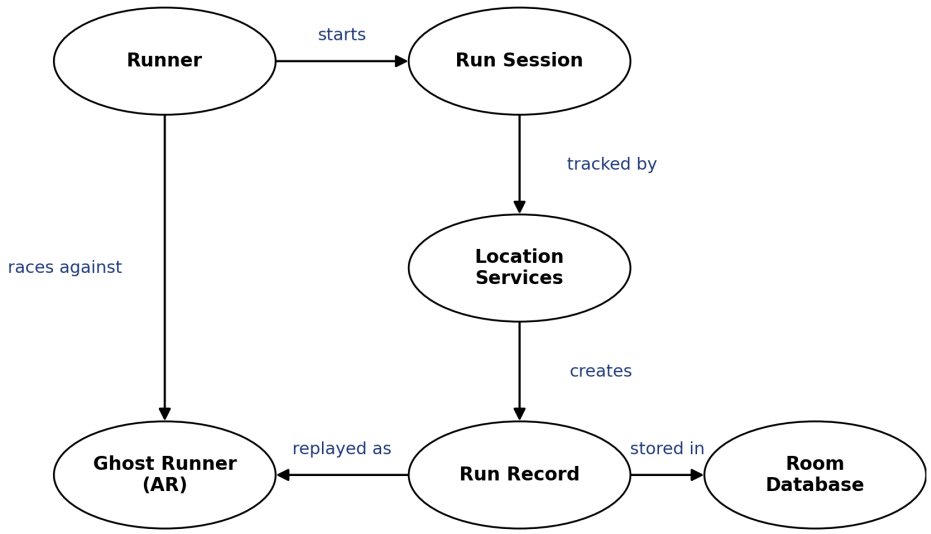


Figure 1.1: Entire work-flow of MirrorWalk

1.7 References

The sources used in this document are listed in the References chapter at the end.

2 Overall Description

2.1 Product Perspective

MirrorWalk is a new, self-contained, native Android application. It is not a part of a bigger system, but it depends on several external components that are provided by Android and by open-source libraries:

- the **Fused Location Provider** (Android system) for location updates [4];
- the **Room database** (local storage) for saving runs [3];
- **MapLibre with OpenFreeMap tiles** for showing maps, with no Google Maps key or billing account [7, 8];
- **ARCore** for understanding the real world through the camera [1], and **SceneView with Filament** for drawing the 3D ghost runner [5, 6].

The users and these external systems are shown in Figure 2.2.

2.2 Product Functions

The main functions of MirrorWalk are:

1. Start, pause, resume and stop a run or walk session.
2. Request location permission when a run is started.
3. Track the GPS location continuously and filter bad points.
4. Show live distance, pace and elapsed time.
5. Draw the route live on a map and show the final route.
6. Save completed runs locally and show a run history.
7. Open a saved route on the map again, or delete it.
8. Switch on Ghost Mode before a session.
9. Show an AR ghost runner anchored to the ground that moves according to the recorded timestamps.
10. Compare the live user's distance and pace with the ghost's.

11. [**Planned**] Profiles, followers/following, posts with run photos, and a social feed.
12. [**Planned**] Fitness analysis.

2.3 Data Model

Figure 2.1 shows the main data stored by MirrorWalk. A `Run` has many `GPS_Point` records (latitude, longitude, timestamp) and can be used as the source of a `Ghost_Run`. These three are stored on the device using Room. The `Users`, `Post` and `Comment` tables belong to the planned social release.

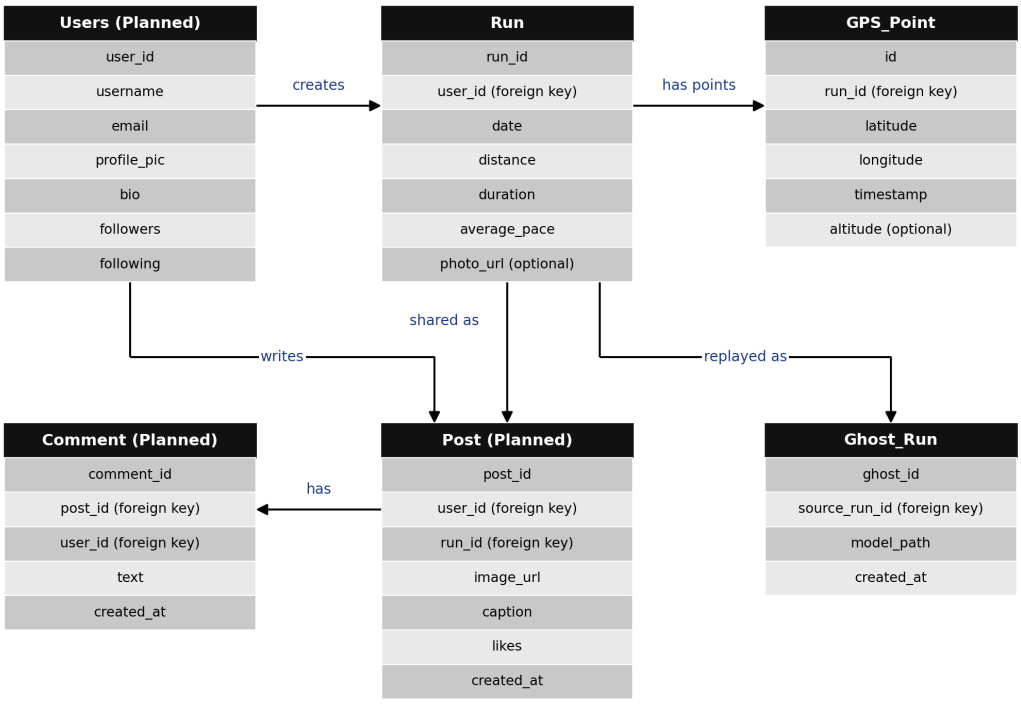


Figure 2.1: Data model of MirrorWalk

2.4 User Classes and Characteristics

User class	Characteristics
Runner (primary user)	A person who runs or walks for fitness. May be a beginner or experienced. Needs a simple screen that is easy to read while moving. Uses all tracking and ghost features.
Follower / Social user **[Planned]**	A runner who views other runners' profiles and posts, and follows or unfollows them.
External systems (actors)	Fused Location Provider, Room database, MapLibre/OpenFreeMap, ARCore and SceneView/Filament. They provide services to the app.

Figure 2.2 shows the types of users and the external systems that work with MirrorWalk.

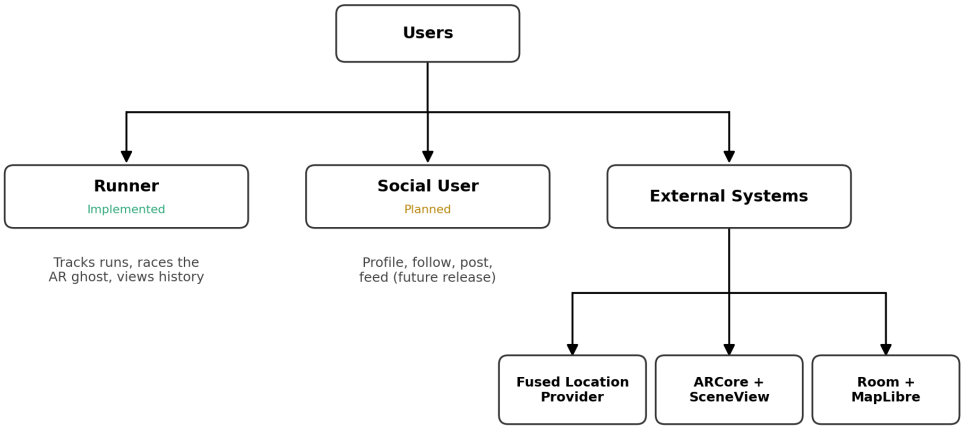


Figure 2.2: Types of users of MirrorWalk

2.5 Operating Environment

- Platform: Android smartphones. The device must support ARCore for the ghost-runner feature.
- Required hardware: GPS, camera and motion sensors.
- Required permissions: fine location, coarse location, camera and internet access.
- Internet is needed only to load map tiles. Run tracking and saved runs work with local data.

2.6 Design and Implementation Constraints

- Language: Kotlin 2.4.x; Java runtime JDK 17.
- IDE and build: Android Studio Quail 4 (2026.1.4), Gradle 9.6, Android Gradle Plugin 9.4.0.
- UI: Jetpack Compose with Material 3 [2] and Navigation Compose [10].
- State management: Kotlin Coroutines [12], StateFlow and Android ViewModel [11].
- Local database: Room 2.8.5. AR: ARCore 1.56.0. 3D: SceneView AR 4.34.0 with Filament.
- No paid map service may be used (MapLibre with OpenFreeMap is used instead).
- Visual design: dark navy/charcoal theme with mint/lime accents, rounded cards, bottom navigation and a central action button. Manrope is the main font and Space Grotesk Bold is used for subtitles.

2.7 Assumptions and Dependencies

- The user's phone has working GPS and the user grants the required permissions.
- The phone supports ARCore and has a camera that works.
- The user runs in an area with reasonable GPS signal.
- Map tiles come from the OpenFreeMap service, so the map background needs internet.
- The ghost run uses hardcoded sample data in release 1.0. The data format is the same as a normal saved run so that it can be replaced later.

2.8 User Activities

2.8.1 Runner Activities

A runner first grants location permission, optionally chooses Ghost Mode, and starts the run. During the run the app tracks and filters GPS points and shows live distance, pace and time. When the runner stops, the run is saved and appears in the Run History, where it can be opened on the map or deleted (Figure 2.3).

2.8.2 Ghost Runner Activities

In Ghost Mode the runner selects a previous run (or sample data). The AR system starts a session, detects the ground, places the 3D ghost runner and replays the recorded timestamps, while the screen compares the runner's live stats with the ghost's (Figure 2.4).

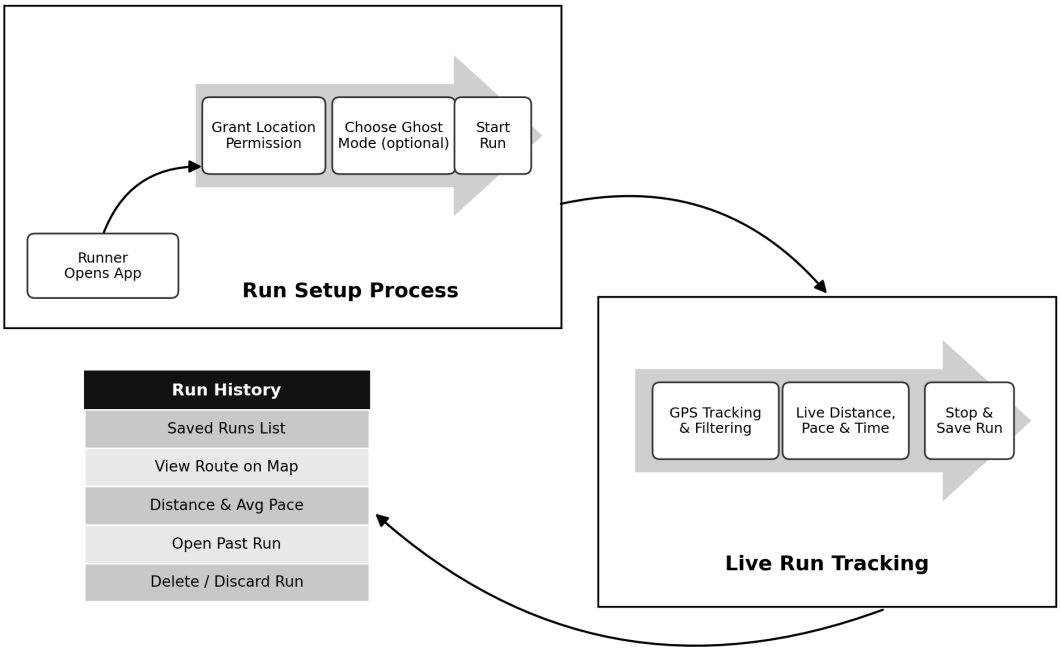


Figure 2.3: Runner activities in MirrorWalk

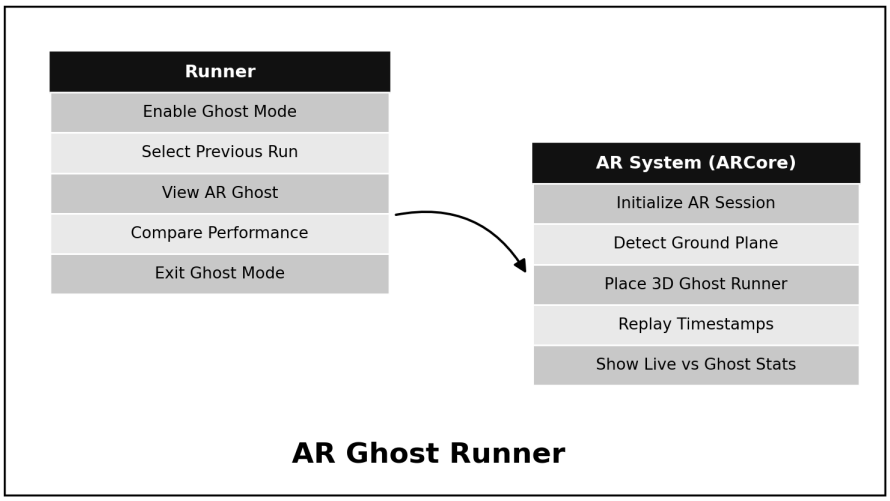


Figure 2.4: Ghost-runner activities in MirrorWalk

2.8.3 Social Activities (Planned)

Social users will be able to sign up, edit a profile, follow other runners, post a run with an optional photo and view a feed (Figure 2.5). These activities are not part of release 1.0.

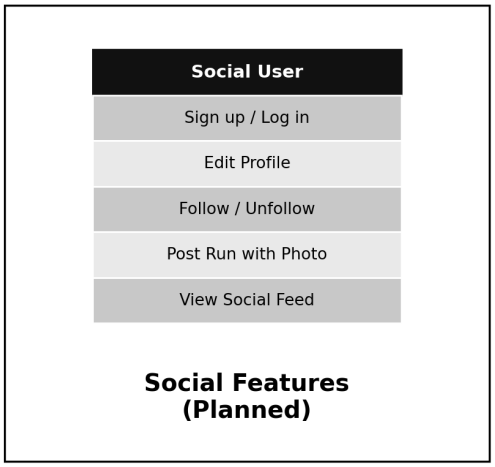


Figure 2.5: Social activities in MirrorWalk (planned)

3 External Interface Requirements

3.1 User Interfaces

- The app shall use a dark navy/charcoal background with mint/lime accent colours, rounded cards, a bottom navigation bar and a central action button for starting a run.
- The main screens are: Run screen, Run History screen, Route view (map), and AR Ghost Run screen.
- The Run screen shall show elapsed time, distance and pace in large, readable text, together with Start, Pause, Resume and Stop controls and a Ghost Mode switch.
- The AR screen shall show the camera view with the ghost runner and an overlay comparing the user and the ghost.
- Text shall use Manrope as the main font and Space Grotesk Bold for subtitles.

3.2 Hardware Interfaces

- **GPS / location hardware** for position updates.
- **Camera** for AR.
- **Motion sensors** used by ARCore to track the phone's movement.
- **Touch screen** for all user input.

3.3 Software Interfaces

Component	**Purpose**
Fused Location Provider	Gives latitude, longitude and timestamp during a session.
Room 2.8.5	Stores runs and their GPS points on the device.
MapLibre Native SDK + OpenFreeMap	Draws the map and the route.
ARCore 1.56.0	Detects planes and tracks the camera pose.

Component	Purpose
SceneView AR 4.34.0 + Filament	Renders and animates the ghost runner model.

3.4 Communications Interfaces

The only network use in release 1.0 is downloading map tiles from OpenFreeMap over the internet. A server and user accounts will be needed for the planned social features and are outside the current prototype.

4 System Features

This chapter lists the features of MirrorWalk. Each feature has a description, a priority, a status and its functional requirements.

4.1 Run/Walk Session Control

Priority: High **Status:** [Implemented]

The user controls a session with Start, Pause, Resume and Stop.

- `FR-1.1`: The system shall let the user start a session.
- `FR-1.2`: The system shall let the user pause a running session and stop recording points while paused.
- `FR-1.3`: The system shall let the user resume a paused session.
- `FR-1.4`: The system shall let the user stop a session and show the final result.

4.2 Location Permission

Priority: High **Status:** [Implemented]

- `FR-2.1`: When the user starts a run, the system shall request location permission at runtime if it is not already granted.
- `FR-2.2`: The system shall start tracking only after permission is granted, and shall inform the user if it is denied.

4.3 GPS Tracking and Filtering

Priority: High **Status:** [Implemented]

- `FR-3.1`: The system shall track the location continuously during an active session.
- `FR-3.2`: The system shall record latitude, longitude and timestamp for each valid GPS point.
- `FR-3.3`: The system shall reject points with poor accuracy.
- `FR-3.4`: The system shall reject points that imply an unrealistic speed.
- `FR-3.5`: The system shall reduce jitter, so that small movements of the reported location while the user is standing still do not add distance.

4.4 Live Statistics

Priority: High **Status:** [Implemented]

- `FR-4.1`: The system shall calculate and display distance in kilometres, updated live.
- `FR-4.2`: The system shall calculate and display pace in minutes per kilometre, updated live.
- `FR-4.3`: The system shall display elapsed time, updated live.

4.5 Route Display

Priority: High **Status:** [Implemented]

- `FR-5.1`: The system shall draw the route on the map as the user moves.
- `FR-5.2`: The system shall display the final route after the run is stopped.

4.6 Saving Runs

Priority: High **Status:** [Implemented]

- `FR-6.1`: The system shall save a completed run locally, including route points, total distance and average pace.

4.7 Run History

Priority: Medium **Status:** [Implemented]

- `FR-7.1`: The system shall provide a Saved Runs screen that lists previous runs.
- `FR-7.2`: The system shall let the user open any saved run and view its route on a map.
- `FR-7.3`: The system shall let the user delete or discard a saved run.

4.8 Ghost Mode and AR Ghost Runner

Priority: High **Status:** [Implemented as prototype]

This is the main feature of MirrorWalk.

- `FR-8.1`: The system shall provide a Ghost Mode switch before a session starts.

- FR-8.2: The system shall load ghost route data from a source structured like a saved run. In release 1.0 this is hardcoded sample data, replaceable later by a stored run.
- FR-8.3: The system shall open an AR screen showing a translucent placeholder runner.
- FR-8.4: The system shall detect a ground plane using ARCore and anchor the ghost runner to it.
- FR-8.5: The system shall move the ghost based on the recorded timestamps of the ghost run.

4.9 Live Comparison Overlay

Priority: High **Status:** [Implemented]

- FR-9.1: The system shall show the user's live distance and pace on the AR screen.
- FR-9.2: The system shall show the ghost's distance and pace at the same moment, next to the user's values.

4.10 Social Features

Priority: Medium **Status:** [Planned]

- FR-10.1: The system shall let users sign up and log in.
- FR-10.2: The system shall provide user profiles with follower and following counts, and let users edit their profile.
- FR-10.3: The system shall let users follow and unfollow other users.
- FR-10.4: The system shall let users post a run with an optional run photo.
- FR-10.5: The system shall show a social feed of posts.

4.11 Fitness Analysis

Priority: Medium **Status:** [Planned]

- FR-11.1: The system shall show fitness analysis based on past runs, such as pace and distance trends.
- FR-11.2: The system shall allow any real saved run (own or a friend's) to be chosen as the ghost run.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- `NFR-1`: Live distance, pace and time shall update smoothly during a run without freezing the screen.
- `NFR-2`: GPS processing shall run in the background (using coroutines) so the UI stays responsive.
- `NFR-3`: The AR ghost shall move smoothly and stay fixed to the ground while the user moves.
- `NFR-4`: The app shall use battery sensibly during long sessions.

5.2 Safety Requirements

- `NFR-5`: The interface shall be simple and use large text so that a user can glance at it quickly while running.
- The AR screen is for use in safe, open areas. The user is responsible for watching their surroundings while running.

5.3 Security and Privacy Requirements

- `NFR-6`: The app shall ask only for the permissions it needs (location, camera, internet).
- `NFR-7`: Saved runs shall stay on the device in release 1.0 and shall not be sent anywhere.
- `NFR-8` [**Planned**]: Login and sharing of runs shall use secure authentication, and users shall control what they share.

5.4 Software Quality Attributes

- **Usability:** simple screens, bottom navigation and a central start button.
- **Reliability:** GPS filtering shall prevent wrong points from spoiling distance and pace. Saved runs shall not be lost when the app is closed.

- **Availability offline:** tracking and run history work without internet.
- **Maintainability:** the code is layered (UI, ViewModel, services, data) so parts can be tested and changed separately.
- **Extensibility:** the ghost data format is the same as a saved run, so real runs can be plugged in later.
- **Portability:** runs on Android devices that support ARCore.

5.5 Business Rules

- MirrorWalk shall not need a paid map service or API key.
- Only valid GPS points shall count towards distance and pace.
- A run shall be saved only when the user stops the session.

6 Other Requirements

6.1 Requirements Status Summary

Feature	**Priority**	**Status**
Session control and live stats	High	Implemented
GPS tracking and filtering	High	Implemented
Route display on map	High	Implemented
Local saving and run history	High / Medium	Implemented
Ghost mode and AR ghost runner	High	Prototype (sample data)
Live comparison overlay	High	Implemented
Profiles, followers, posts, feed	Medium	Planned
Fitness analysis	Medium	Planned
Real saved run as ghost	High	Planned

6.2 Maintenance and Future Changes

MirrorWalk will need regular maintenance, because Android, ARCore and the libraries it uses are updated often. The requirements may also change as the social layer, fitness analysis and real-run ghost features are added. Future work includes better GPS smoothing, battery optimisation for long AR sessions, and testing on more Android devices.

References

[1] Google Developers, "Geospatial developer guide," ARCore Documentation, 2026. https://developers.google.com/ar/develop/geospatial

[2] Android Developers, "Jetpack Compose," 2026. https://developer.android.com/develop/ui/compose

[3] Android Developers, "Save data in a local database using Room," 2026. https://developer.android.com/training/data-storage/room

[4] Google Developers, "LocationServices," Google Play services Documentation, 2026. https://developers.google.com/android/reference/com/google/android/gms/location/LocationServices

[5] SceneView, "SceneView: 3D & AR SDK for Android," GitHub, 2026. https://github.com/SceneView/sceneview

[6] Google, "Filament: Real-time physically based rendering engine," GitHub, 2026. https://github.com/google/filament

[7] MapLibre, "MapLibre Native Android Developer Documentation," 2026. https://maplibre.org/maplibre-native/docs/book/platforms/android/

[8] OpenFreeMap, "OpenFreeMap Quick Start Guide," 2026. https://openfreemap.org/quick_start/

[9] Kotlin, "Kotlin Documentation," JetBrains, 2026. https://kotlinlang.org/docs/home.html

[10] Android Developers, "Navigation with Compose," 2026. https://developer.android.com/develop/ui/compose/navigation

[11] Android Developers, "ViewModel," 2026. https://developer.android.com/topic/libraries/architecture/viewmodel

[12] Android Developers, "Kotlin coroutines on Android," 2026. https://developer.android.com/kotlin/coroutines